

Multi-criterion Analysis of Reference Architectures and Modeling Languages used in Production Systems Modeling

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Abstract—Due to the environment within which they are currently developing, production systems make use of new organizing and operation methods as well as the support which information and communication technology has to offer in order to remain competitive and to survive on the market. Within this context, the present paper describes, in a pluridisciplinary approach and by means of the multi-attribute decision method, the methods of selecting the reference architecture and modeling languages which are perfect candidates for implementation in the process of modeling and integration of the company's compartments and of the company with the upstream and downstream organizations.

Index Terms—multi-attribute analysis, reference architectures, modeling languages.

I. INTRODUCTION

Taking into account the huge possibilities (practically unrestrained) which the Internet and its most dynamic and fascinating service, the WWW offer, finding information is no longer a problem that widens the gap between researchers in developed countries and those in developing countries. The only problem all scientists have to face is the massive amount of information and that of identifying the technology that is appropriate to each society, considering the possibilities which the market offers and the level of automation.

The article presents in a scientific manner, by means of the multi-criterion analysis method, the reference architectures and the modeling languages which are the most suitable for enterprise integration considering the human factor, the most important resource in the process of transition to the EU economy. The researches conducted so far are part of a fundamental research project of the Romanian Academy (phase I), which intends to present these methods and techniques that are the most suitable to Romanian economy; the second phase of this project, which uses the results of the first, implies the modeling and integration of production systems.

II. THE ACTUAL SITUATION IN OUR COUNTRY

The current phase of Romanian enterprises is characterized by a low level of automation and the existence of those automation islands which generate disturbances in information processing, thus preventing managers from making effective decisions at the right time.

III. REFERENCE ARCHITECTURES

The reference architecture (RA), which is used in the process of integration, significantly influences the way in which the company is modeled.

For this reason, the decision method based on the multi-attribute analysis has been employed in the process of the company's modeling and integration.

The process of selecting a reference architecture that is to be used in the process of company's modeling and integration required two phases:

- *the first phase* of the selection process consists in a "survey" of the AR existent on the market. These are sorted according to a limited number of criteria in order to "pick" the most representative RA;
- *the second phase* is a decision based on the multiple-attribute analysis, when the selected RA must be ranked in order to identify the most suitable.

Within this framework and considering the researches presented in the literature [18], [11], [4], [10], [8], [2], it is obvious that the first phase *has identified* – as a result of the sorting process – the following reference architectures: GERAM, ARIS, CIMOSA, GRAI/GIM, IEM and PERA which form the aggregate of reference architectures considered in the second phase $A = \{A_1, A_2, A_3, A_4, A_5, A_6\}$. The correspondence of the elements of aggregate A to RA is the following: $A_1 = \text{GERAM}$, $A_2 = \text{ARIS}$, $A_3 = \text{CIMOSA}$, $A_4 = \text{GRAI/GIM}$, $A_5 = \text{IEM}$ și $A_6 = \text{PERA}$;

The second phase, which is actually the decision-making process based on the multi-attribute analysis, falls into three major activities:

- defining the set of evaluation criteria;
- estimating the value and usability of each architecture;
- applying a ranking method.

A. Defining the Set of Evaluation Criteria

In order to ensure the most appropriate decisions are made in the case of the selection of a reference architecture, several requirements have been taken into account [5] such as: completeness, non-redundancy, decomposability, operability, and the characteristic of being minimum. Thus, there are six evaluation criteria:

- present or future standardizations;
- adaptability to Romanian industry;
- generity size;
- industrial usage;

- life-cycle size;
- model-perspectives size,

which represented in a descending order as $E_1, E_2, \dots, E_6$ forms the criteria aggregate considered in the decision-making process $E=\{E_1, E_2, E_3, E_4, E_5, E_6\}$. The significance of each of these criteria for the decision-making process is dissimilar and thus they have been assigned equivalent weight. The weights aggregate forms the *weight coefficient vector* of the criteria and is represented as follows: $w=\{w_1, w_2, w_3, w_4, w_5, w_6\}$ with the values $w=\{0.30, 0.25, 0.15, 0.10, 0.10, 0.10\}$, on the condition:

$$\left(\sum_{j=1}^{nc} w_j = 1 \right) \quad (1)$$

where: nc represents the number of evaluation criteria.

B. Establishing a Reference Architecture Classification

In order to establish a final classification, one must first prepare a decision table. Decision tables are fundamental input elements for solving multi-attribute decision problems [6]. The initial, unprocessed form is also termed "the consequence table/matrix" and links the alternatives (in this case, the reference architectures) to the evaluation criteria by means of scores.

Table 1 presents the consequence matrix resulted from the analysis of each architecture according to established criteria where the columns represent the criteria aggregate $E=\{E_1, E_2, E_3, E_4, E_5, E_6\}$, and the rows represent the alternatives or the considered aggregate of architectures $A=\{A_1, A_2, A_3, A_4, A_5, A_6\}$.

Thus, as the consequence matrix (table 1) or the diagram in figure 2 show, the classification is the following: alternatives A_3, A_1 and A_2 occupy positions I, II and III; alternatives A_3, A_1 and A_2 correspond to the CIMOSA, GERAM and ARIS reference architectures.

Table 1. The Consequence Matrix for the RA Selection Problem

Alternatives	Evaluation Criteria						Scores	Classification
	E_1	E_2	E_3	E_4	E_5	E_6		
A_1	4	5	5	1	5	5	25	II
A_2	1	3	5	4	3	5	21	III
A_3	5	3	5	5	4	5	27	I
A_4	1	3	3	2	3	5	17	V
A_5	1	2	5	2	4	3	17	V

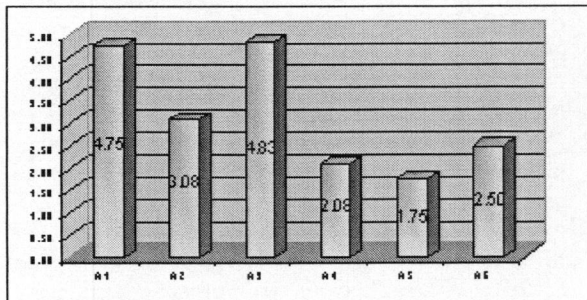


Fig. 1. RA Classification Resulted from the Consequence Table

In order to extract the decision matrix (table 2) from the consequence matrix (table 1) the *weighted additive function method*. Thus, for each alternative A_i ($i=1,2,\dots,na$), the performance indicator is determined or the aggregate value function by means of the following expression:

$$J_i = \sum_{j=1}^{nc} w_j r_j \quad (2)$$

where: w is the weights vector w_j of the criteria E_j ($j=1,2,\dots,nc$), nc is the number of evaluation criteria, and r_j represents the scores normalized according to the following formulae:

$$r_{ij} = 1 - \frac{(s_{Mj} - s_{ij})}{\Delta s_j}, \text{ if } E_i \text{ is the minimum criterion,} \quad (3)$$

$$r_{ij} = 1 - \frac{(s_{ij} - s_{Mj})}{\Delta s_j}, \text{ if } E_i \text{ is the maximum criterion,} \quad (4)$$

where Δs_j is the variation range defined by the extreme scores (s_{mj} and s_{Mj}) of the considered alternatives and is calculated by means of formula number 5.

$$\Delta s_j = s_{Mj} - s_{mj} \quad (5)$$

The pairs (s_{mj} and s_{Mj}) of the lowest values and, respectively, of the highest from the viewpoint of the E_j criterion are:

$$s_{mj} = \min(s_{ij}, i = 1,2,\dots,na); j = 1,2,\dots,nc, \quad (6)$$

$$s_{Mj} = \max(s_{ij}, i = 1,2,\dots,na); j = 1,2,\dots,nc, \quad (7)$$

where: na is the number of alternatives, and nc represents the number of evaluation criteria.

Table 2. The Decision Table for the RA Selection Problem

Alternatives	Evaluation Criteria					
	$As_1=4$	$As_2=3$	$As_3=4$	$As_4=4$	$As_5=2$	$As_6=2$
	E_1	E_2	E_3	E_4	E_5	E_6
A_1	0.75	1.00	1.00	0.00	1.00	1.00
A_2	0.00	0.33	1.00	0.75	0.00	1.00
A_3	1.00	0.33	1.00	1.00	0.50	1.00
A_4	0.00	0.33	0.50	0.25	0.00	1.00
A_5	0.00	0.00	1.00	0.25	0.50	0.00
A_6	0.00	1.00	0.00	0.50	0.50	0.50
Weights	0.30	0.25	0.15	0.10	0.10	0.10

The decision matrix number 2 is obtained by normalizing the scores in the consequence matrix (table 1) by using formulae 3, 4, 5, 6 and 7.

In order to establish a final score and a classification of the six alternatives formula 2 is used and table 3 is thus obtained.

Table 3. The Final Score and RA Classification by the Multi-Attribute Decision Analysis

	Final Score	Final Classification
A ₁ – GERAM	0.83	I
A ₂ – ARIS	0.41	III
A ₃ – CIMOSA	0.78	II
A ₄ – GRAI/GIM	0.28	V
A ₅ – IEM	0.23	VI
A ₆ – PERA	0.40	IV

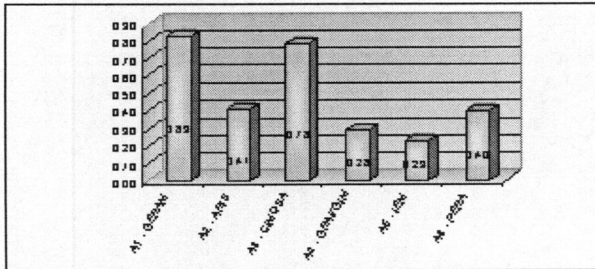


Fig. 2. The Classification Obtained by the Multi-Attribute Decision Analysis

Thus, the first place is occupied by the GERAM reference architecture, the second place by CIMOSA, and the third place by ARIS.

The comparative results obtained on the basis of the multi-attribute decision analysis presented above, enable us to draw the following conclusions:

- the most complex reference architecture is GERAM, which, as it is well-known, is a combination of CIMOSA, PERA, GRAI/GIM;
- GERAM covers most of the established comparing criteria; the research that has been conducted so far by ISO committees in support of GERAM development will eventually lead to the development of a standard architecture, which will represent the basis of further research on enterprise development.

The GERAM architecture should be the basis for all the research that will be conducted in the field of integrated production systems. Even if the working environment and general concepts have been defined for this architecture, many steps should be taken with a view to filling the architecture's component "blocks".

IV. MODELLING LANGUAGES

For the comparison of modeling languages the same method has been used, namely that based upon the multi-criterion analysis presented above.

Thus, in order to identify the modeling language which is the most suitable to use in the modeling process, the following modeling languages have been assessed on the basis of the most comprehensive reference material [16], [18], [7], [9], [13], [11], [10], [15], [8], [17], [12], [3], [1], [2], GRAI/GIM, Petri Networks, OOA/OMT, CIMOSA, IDEF0, IDEF1X, IDEF3, IEM, ARIS, EXPRESS and UML, which form the alternatives aggregate $A = \{A_1, \dots, A_{11}\}$.

Thus, 10 criteria have been defined which from the criteria aggregate $E = \{E_1, E_2, \dots, E_{10}\}$:

- from the viewpoint of the *modeling and simulation support*:
 - E₁ = CASE tools;
 - E₂ = methodological support;
- from the viewpoint of *modeling perspectives*:
 - E₃ = the function perspective;
 - E₄ = the information perspective;
 - E₅ = the resource perspective;
 - E₇ = the organization perspective;
- from the viewpoint of the *modeling and coverage principles*:
 - E₇ = problem partition;
 - E₈ = functional decomposition;
 - E₉ = generity;
 - E₁₀ = function partition;

The weight for each criterion is $\{0.20, 0.20, 0.10, 0.10, 0.10, 0.10, 0.05, 0.05, 0.05, 0.05\}$, where the weights aggregate form the vector w , which meets the condition in equation 1, namely the sum of all weights is equal to 1, and thus the decision table is obtained (table 4).

The graphic representation of the results obtained in table 4 are presented in figure 3.

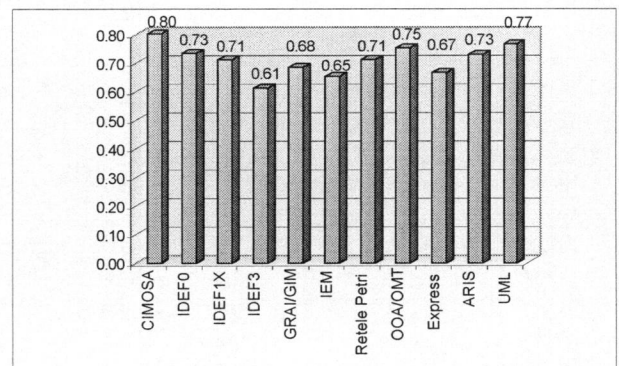


Fig. 3. The Modeling Language Classification Resulted From the Decision Table

As shown in figure 3, the CIMOSA has the best score.

However, the UML is recommended because it is the most suitable language for enterprise modeling, due to the following reasons:

- as shown in table 4, the UML's weaknesses reside in the modeling perspective and the modeling and coverage principles; however, this "drawback" can be overcome if, both the modeling language, and the GERAM enterprise integration methodology, obtained as well on the basis of a multi-criterion analysis, are recommended for the development of a model;
- according to various evaluations [8] the most prevalent methods in enterprise modeling are the object-oriented ones, such as the UML followed by the process-oriented ones, comprised in CIMOSA, ARIS and IEM;
- UML's user-friendliness in terms of its integration with Web technologies which support the development of the desired model. Thus, the data can be modeled by

Table 4. The Decision Table for the Modeling Language Selection

Criteria Alternatives	E ₁	E ₂	E ₃	E ₄	E ₅	E ₆	E ₇	E ₈	E ₉	E ₁₀	Final Score	Final Classification
	$\Delta_1=2$	$\Delta_2=3$	$\Delta_3=3$	$\Delta_4=3$	$\Delta_5=3$	$\Delta_6=3$	$\Delta_7=3$	$\Delta_8=3$	$\Delta_9=3$	$\Delta_{10}=4$		
A ₁	0.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.80	I
A ₂	1.00	1.00	0.00	0.67	0.67	1.00	0.00	0.00	1.00	1.00	0.73	IV
A ₃	1.00	1.00	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.25	0.71	V
A ₄	1.00	1.00	0.00	0.00	0.00	1.00	1.00	0.00	1.00	0.25	0.61	IX
A ₅	0.00	1.00	1.00	0.67	1.00	1.00	0.33	0.00	1.00	1.00	0.68	VI
A ₆	0.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00	0.00	0.65	VIII
A ₇	1.00	0.67	0.00	0.67	1.00	1.00	1.00	0.00	1.00	0.25	0.71	V
A ₈	1.00	1.00	0.00	0.00	1.00	1.00	0.33	1.00	0.67	1.00	0.75	III
A ₉	1.00	1.00	0.00	0.00	1.00	0.00	0.67	1.00	0.67	1.00	0.67	VII
A ₁₀	1.00	0.00	1.00	0.67	1.00	1.00	1.00	1.00	1.00	0.25	0.73	IV
A ₁₁	1.00	1.00	0.00	0.00	1.00	1.00	0.67	1.00	0.67	1.00	0.77	II
Weights	0.20	0.20	0.10	0.10	0.10	0.10	0.05	0.05	0.05	0.05		

means of the UML language and then easily handled by means of the XML meta-language. Moreover, another positive element is the UML's capacity of generating Java codes.

- the fact that computer scientists who deal with industrial systems analysis and design "are into" UML, unlike system analysts and designers who probably prefer IDEF or ARIS.

V. CONCLUSIONS

To conclude, considering the current conditions, we can state that the integration of all enterprise compartments and of the enterprise with its partners, suppliers and clients becomes attainable only through a scientific approach on the selection of methods and technologies used in the complex process of enterprise modeling. In addition, the existing standards have been considered, and in this respect we maintain that the GERAM architecture is the most likely to be standardized, thus overcoming CIMOSA which is currently standardized only at a European level, and the fact that the UML language is standardized (thus offering an easy integration with web technologies which play an active role in the development of the integrated enterprise model) offers even more support for the research presented in this paper.

VI. REFERENCES

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